

**THIRD SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

(Multiple Choice Questions for SDE Candidates)

1. $\int \frac{\ln x}{x} dx = \text{_____}.$

(A) $\frac{1}{x^2} + c.$

(B) $\frac{(\ln x)^2}{2} + c.$

(C) $\frac{\ln x}{2} + c.$

(D) $\frac{1}{x} - \ln x + c$

2. $\int \tan x \, dx = \text{_____}.$

(A) $\ln |\cos x| + c.$

(B) $\ln |\sec x| + c.$

(C) $\sec^2 x + c.$

(D) $\ln |\sin x| + c.$

3. $\lim_{x \rightarrow 1} \frac{1-x}{\ln x} = \text{_____}.$

(A) $-1.$

(B) $+1.$

(C) $0.$

(D) $2.$

4. Range of $\tan^{-1} x$ is _____.

(A) $[-1, 1].$

(B) $(-1, 1).$

(C) $(-\infty, \infty).$

(D) $(0, \infty).$

5. $a_n = \frac{2n+1}{3n+5}$. The $\lim_{n \rightarrow \infty} a_n = \text{_____}.$

(A) $\frac{1}{5}.$

(B) $\frac{2}{5}.$

(C) $\frac{2}{3}.$

(D) $\frac{3}{2}.$

6. The sequence $\left\{ \frac{1}{n} \right\}$ is :

(A) Diverges.

(B) Increasing.

(C) Decreasing.

(D) None of these.

7. $\lim_{n \rightarrow \infty} a_n = 2, \lim_{n \rightarrow \infty} b_n = 4$, then $\lim_{n \rightarrow \infty} (a_n + b_n) = \underline{\hspace{2cm}}$.
- (A) 6. (B) 10.
(C) 8. (D) 16.
8. $\lim_{n \rightarrow \infty} x^{1/n} (x > 0) = \underline{\hspace{2cm}}$.
- (A) 0. (B) 1.
(C) ∞ . (D) None of these.
9. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \underline{\hspace{2cm}}$.
- (A) 1. (B) 0.
(C) e . (D) $\frac{1}{e}$.
10. Fourth term of the sequence $\left\{ \frac{(-1)^{n+1}}{2^{n+1}} \right\}$ is :
- (A) $\frac{1}{2^5}$. (B) $\frac{-1}{2^5}$.
(C) $\frac{1}{2^4}$. (D) $\frac{-1}{2^4}$.
11. The series $\sum_{n=1}^{\infty} a_n$ converges, then :
- (A) $\lim_{n \rightarrow \infty} a_n = c$, a constant. (B) $\lim_{n \rightarrow \infty} a_n = 0$.
(C) $\lim_{n \rightarrow \infty} a_n = \infty$. (D) $\lim_{n \rightarrow \infty} a_n$ does not exist.
12. Let $a_n > 0$ and $b_n > 0$ and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l$. Then $\sum a_n$ and $\sum b_n$ converges if :
- (A) $l = 0$. (B) $l > 0$.
(C) $l = \infty$. (D) None of these.
13. $\sum a_n$ be a series such with $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l$. Then :
- (A) $\sum a_n$ converges if $l > 1$. (B) $\sum a_n$ diverges if $l > 0$.
(C) $\sum a_n$ converges if $l < 1$. (D) $\sum a_n$ diverges if $l = 0$.

14. $\sum a_n$ be a series such with $a_n \geq 0$ $\lim_{n \rightarrow \infty} (a_n)^{1/n} = l$. Then :

- (A) $\sum a_n$ converges if $l < 1$. (B) $\sum a_n$ diverges if $l < 1$.
(C) $\sum a_n$ converges if $l = 1$. (D) $\sum a_n$ diverges if $l = 1$.

15. If the series $\sum a_n (x - c)^n$ converges for $x = k$. Then converges in :

- (A) $|r - c| < k$. (B) $|r - c| \leq k$.
(C) $|r - c| > k$. (D) $|r - c| \geq k$.

16. The series $\sum_{n=0}^{\infty} x^n$ is :

- (A) Converges absolutely for $|x| < 1$. (B) Converges for $|x| > 1$.
(C) Has radius of converges $\frac{1}{2}$. (D) None of these.

17. The vertex of the parabola whose focus is $(1, -1)$ and whose directrix passes through $(3, 3)$ is :

- (A) $(-2, 1)$. (B) $(2, 1)$.
(C) $(1, 2)$. (D) $(-1, 2)$.

18. Equation of the hyperbola with foci $(0, \pm\sqrt{2})$ and asymptote $y = x$:

- (A) $x^2 - y^2 = 1$. (B) $\frac{x^2}{4} - \frac{y^2}{2} = 1$.
(C) $y^2 - x^2 = 1$. (D) $\frac{y^2}{4} - \frac{x^2}{2} = 1$.

19. Eccentricity of the hyperbola $9x^2 - 16y^2 = 144$ is :

- (A) $\frac{7}{5}$. (B) $\frac{5}{4}$.
(C) $\frac{6}{5}$. (D) $\frac{7}{4}$.

20. The equation $Ax^2 + Cy^2 + Dx + Ey + F = 0$ represent a ellipse if :

- (A) $(A > 0 \text{ and } C > 0)$. (B) $(A > 0 \text{ and } C < 0)$.
(C) $(A < 0 \text{ and } C > 0)$. (D) $(A \neq 0, C \neq 0)$.

**THIRD SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

Time : Three Hours

Maximum : 80 Marks

Part A

*Answer all the twelve questions.
Each question carries 1 mark.*

1. Evaluate $\int_0^{\pi/6} \tan 2x \, dx$.
2. Define an alternating series.
3. Find the vertices of the hyperbola $\frac{y^2}{4} - \frac{x^2}{5} = 1$.
4. Find the Taylor polynomial of order zero generated by $f(x) = \sin x$ at $a = \frac{\pi}{4}$.
5. Evaluate $\frac{d}{dt} \left(\tan^{-1} \sqrt{1+t^2} \right)$.
6. Examine whether $\sum_{n=1}^{\infty} n^2$ converges or diverges.
7. Find the directive of the parabola $y^2 = 10x$.
8. Define absolute convergence.
9. Find y if $\ln y = 3t + 5$.
10. Find the parametric equation of the circle $x^2 + y^2 = a^2$.

Turn over

11. Examine whether $x^2 + xy + y^2 - 1 = 0$ represents a parabola, ellipse or hyperbola.

12. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$.

(12 × 1 = 12 marks)

Part B

Answer any **nine** questions.
Each question carries 2 marks.

13. For what values of x do the series $\sum_{n=0}^{\infty} n! x^n$ converges.

14. Evaluate $\frac{d}{dx} \ln_{10}(3x+1)$.

15. Evaluate $\int \frac{\log_2 x}{x} dx$.

16. Find the Taylor series generated by $f(x) = \frac{1}{x}$ at $a = 2$.

17. Find $\frac{dy}{dx}$ if $y = x^x, x > 0$.

18. Graph the set of points whose polar co-ordinates satisfy the conditions $1 \leq r \leq 2$ and $0 \leq \theta \leq \frac{\pi}{2}$.

19. Examine whether the series :

$5 + \frac{2}{3} + 1 + \frac{1}{7} + \frac{1}{2} + \frac{1}{3!} + \frac{1}{4!} + \dots + \frac{1}{k!} + \dots$ converges.

20. Evaluate $\int_0^{\ln 2} 4e^x \sin hx \, dx$.

21. Show that $\ln x$ grows slower than x as $x \rightarrow \infty$.

22. Examine whether $\sum_{n=1}^{\infty} (-1)^{n+1}$ converges or diverges.

23. Evaluate $\lim_{x \rightarrow 0} \frac{3x - \sin x}{x}$.

24. Evaluate $\int_{-\pi/2}^{\pi/2} \frac{4 \cos \theta}{3 + 2 \sin \theta} d\theta$.

(9 × 2 = 18 marks)

Part C

Answer any **six** questions.
Each question carries 5 marks.

25. The standard parametrization of the circle of radius 1 centered at the point (0, 1) in the xy -plane is $x = \cos t$, $y = 1 + \sin t$, $0 \leq t \leq 2\pi$. Use the parametrization to find the area of the surface swept out by revolving the circle about the x -axis.

26. Find the centroid of the first quadrant of the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq 2\pi$.

27. Show that $\lim_{x \rightarrow 0^+} (1+x)^{\frac{1}{x}} = e$.

28. Using Integral test show that the p -series :

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \dots + \frac{1}{n^p} + \dots \text{ converges if } p > 1 \text{ and diverges if } p \leq 1.$$

29. Find the Taylor polynomial generated by $f(x) = \cos x$ at $x = 0$.

30. Find the length of the Cardioid $r = 1 - \cos \theta$.

31. Prove that if $\sum_{n=1}^{\infty} |a_n|$ converges then $\sum_{n=1}^{\infty} a_n$ converges.

32. Graph the curve $r^2 = 4 \cos \theta$.

33. Investigate the convergence of the series $\sum_{n=1}^{\infty} \frac{(2n)!}{n! n!}$.

(6 × 5 = 30 marks)

Part D

*Answer any two questions.
Each question carries 10 marks.*

34. Find the area of the region in the plane enclosed by the Cardioid $r = 2(1 + \cos \theta)$.

35. Show that the Maclaurin's series for $\cos x$ converges to $\cos x$ for every value of x .

36. Using Integral test examine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

(2 × 10 = 20 marks)